

Lesson 06 Demo 03

Creating Azure Front Door Service

Objective: To create an Azure Front Door service, ensuring secure and reliable access to applications worldwide

Tools Required: Azure account with administrator access

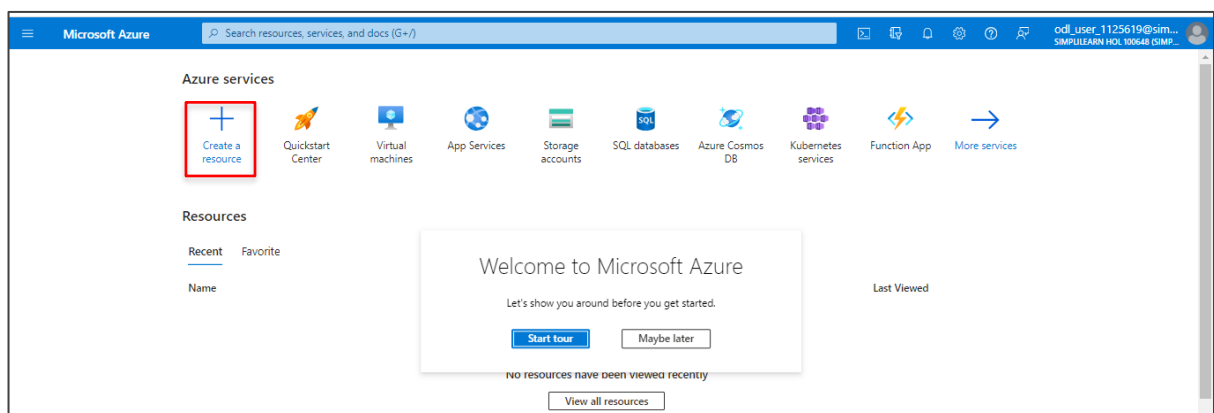
Prerequisites: Two Web Apps

Steps to be followed:

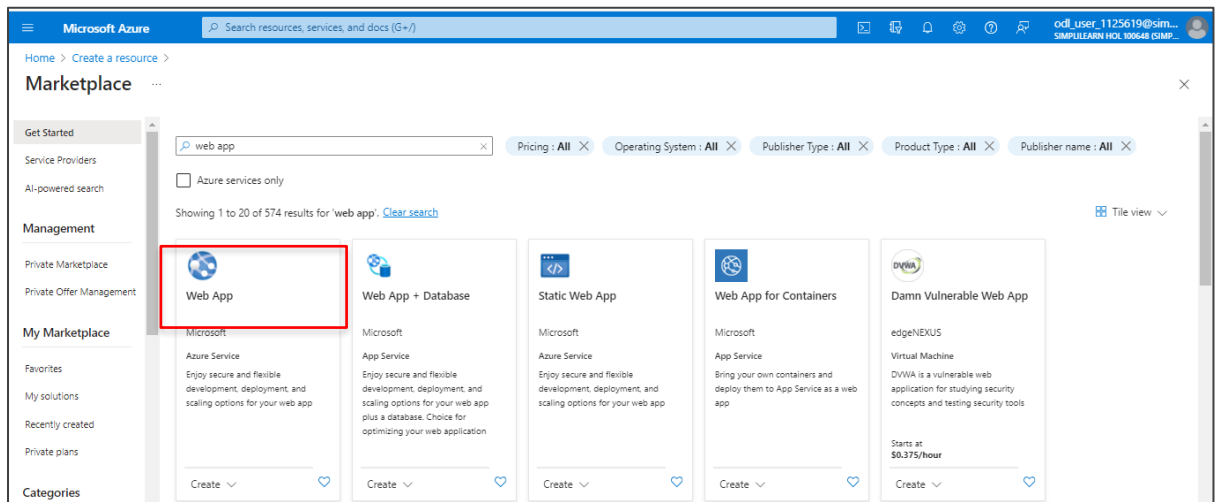
1. Create a web app
2. Configure the Azure Front Door

Step 1: Create a web app

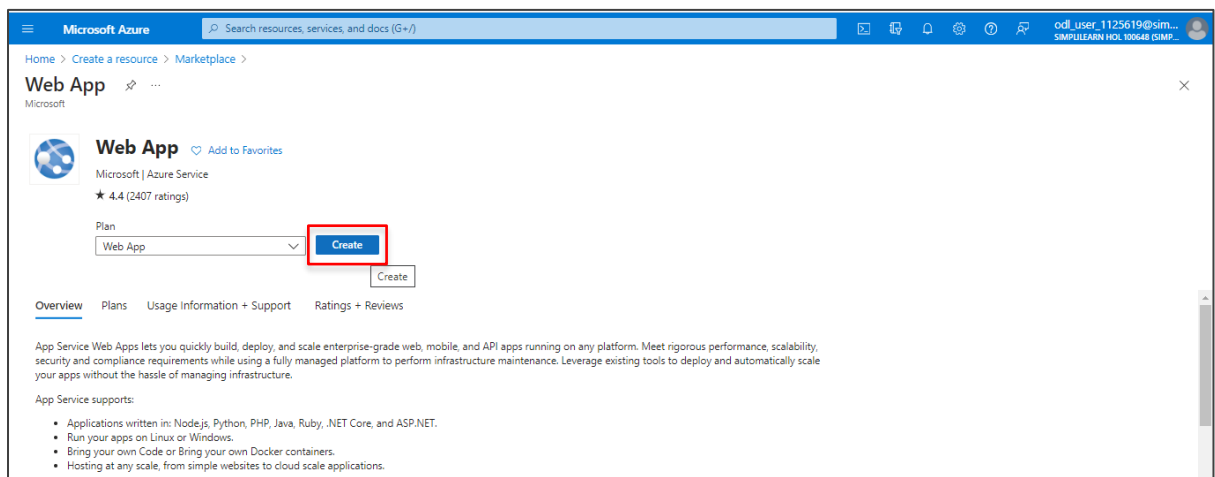
1.1 Log in to the Azure portal and click on **+ Create a resource**



1.2 In the **Marketplace** page, under the **Get Started** menu, search for and select **Web App**



1.3 Click on **Create**



1.4 In the **Create Web App** page, under the **Basics** tab, create a Resource Group (**rg01**)

Home > Create a resource > Marketplace > Web App >

Create Web App

Basics Deployment Networking Monitoring Tags Review + create

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group * [Create new](#)

Instance Details

Need a database? Try the new Web + Database experience.

Name *

Publish * ☐ Code ☐ Docker Container ☐ Static Web App

Runtime stack *

A resource group is a container that holds related resources for an Azure solution.

1.5 Under **Instance Details**, enter the **Name** as **mydemowebapp11** and select **Runtime stack** as **.Net 6(LTS)**

Home > Create a resource > Marketplace > Web App >

Create Web App

Basics Deployment Networking Monitoring Tags Review + create

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

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Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group * [Create new](#)

Instance Details

Need a database? Try the new Web + Database experience.

Name *

Publish * ☒ Code ☐ Docker Container ☐ Static Web App

Runtime stack *

1.6 Under **Operating System**, choose **Windows**, select **Pricing plan** as **Free F1(Shared infrastructure)**, and click on **Review + create**

Microsoft Azure

Home > Create a resource > Marketplace > Web App

Create Web App

Operating System * ☐ Linux ☒ Windows

Region *
Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans
 App Service plan pricing tier determines the location, features, cost and compute resources associated with your app. [Learn more](#)

Windows Plan (East US) *
Create new

Pricing plan
Explore pricing plans

Zone redundancy
 An App Service plan can be deployed as a zone redundant service in the regions that support it. This is a deployment time only decision. You can't make an App Service plan zone redundant after it has been deployed. [Learn more](#)

Zone redundancy ☐ Enabled: Your App Service plan and the apps in it will be zone redundant. The minimum App Service plan instance count will be three.
☒ Disabled: Your App Service Plan and the apps in it will not be zone redundant. The minimum App Service plan instance count will be one.

[Review + create](#) [< Previous](#) [Next: Deployment >](#)

1.7 Click on **Create**

Microsoft Azure

Home > Create a resource > Marketplace > Web App

Create Web App

Basics Deployment Networking Monitoring Tags Review + create

Summary

Web App by Microsoft **Free sku**
 Estimated price - Free

Details

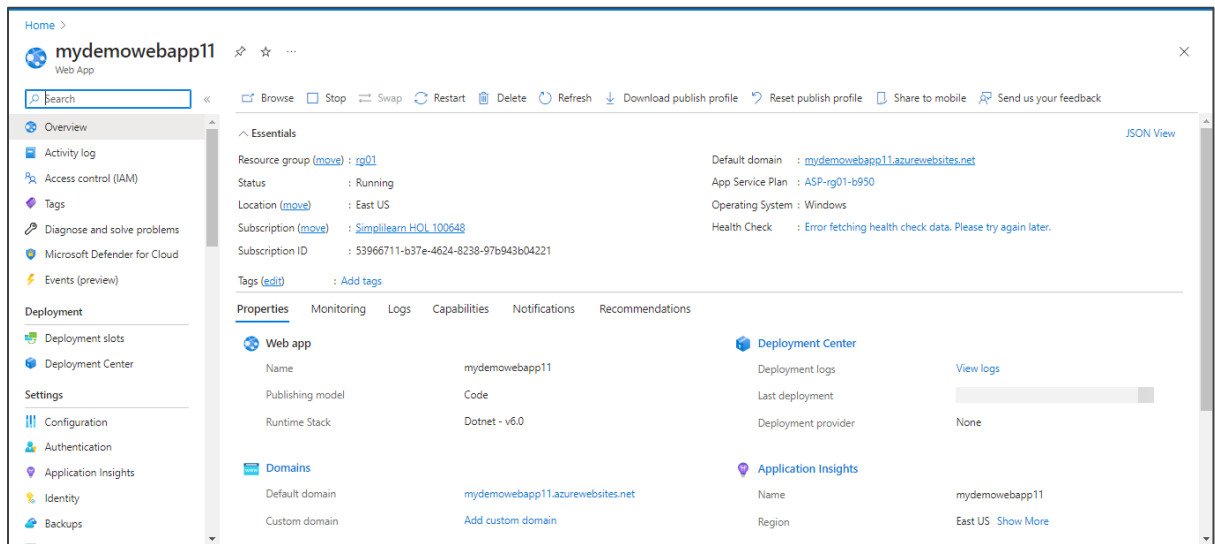
Subscription 53966711-b37e-4624-8238-97b943b04221
 Resource Group rg01
 Name mydemowebapp11
 Publish Code
 Runtime stack .NET 6 (LTS)

App Service Plan (New)

Name ASP-rg01-b950
 Operating System Windows
 Region East US
 SKU Free
 ACU Shared infrastructure

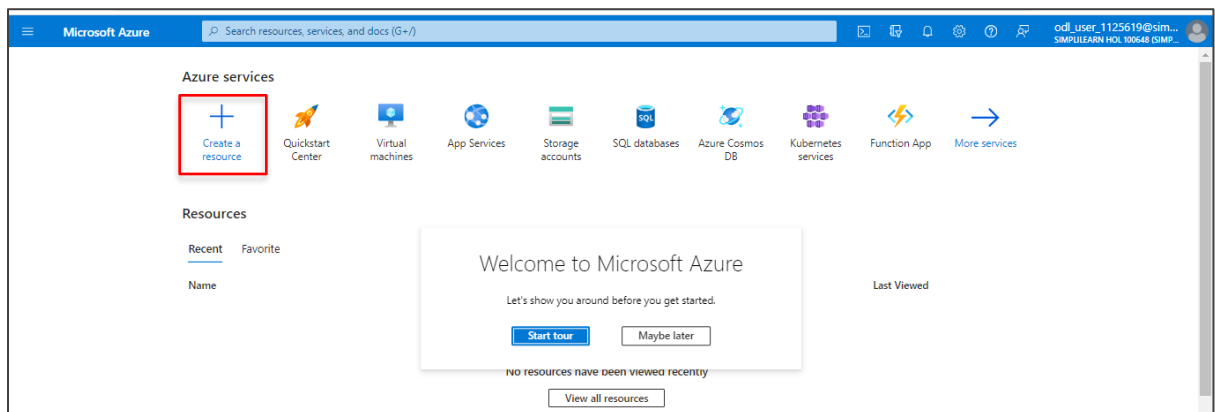
[Create](#) [< Previous](#) [Next >](#) [Download a template for automation](#)

The output will appear as follows:

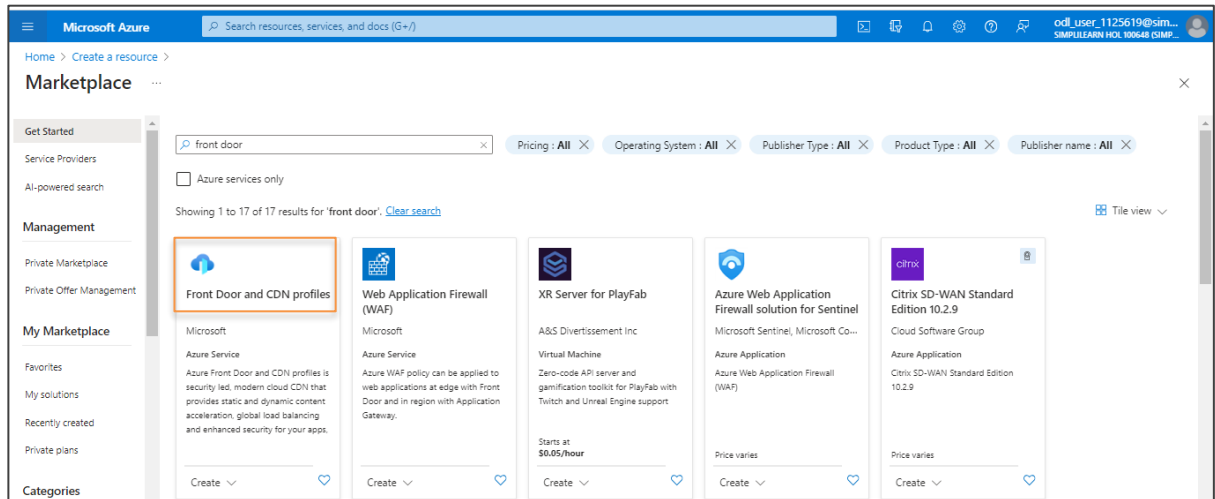


Step 2: Configure the Azure Front Door

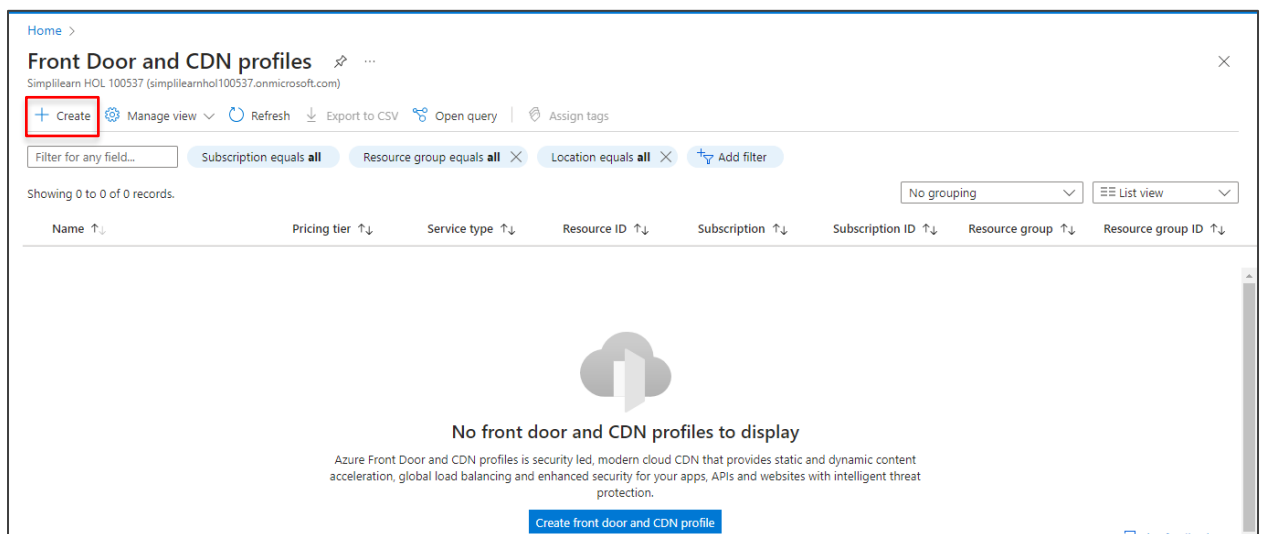
2.1 Log in to the Azure portal and click on + Create a resource



2.2 On the **Marketplace** page, under the **Get Started** menu, search and select **Front Door** and **CDN** profiles



2.3 Click on **+ Create**



2.4 Under **Compare offerings**, select **Custom create**

Home > Create a resource > Marketplace > Front Door and CDN profiles >

Compare offerings

Microsoft Azure

Choose between Azure Front Door and other offerings.

Azure Front Door

Azure Front Door is a secure cloud CDN which provides static and dynamic content acceleration, global load balancing and protection of your apps, APIs and websites with intelligent threat protection.

Explore other offerings

See offerings for our Azure Front Door (classic) and Azure CDN Standard from Microsoft (classic), along with our partner offerings.

Choose between Azure Front Door options

Quick create

Get started with a simplified web application deployment using default settings.

Define one endpoint with one origin and one WAF policy to get your Front Door up and running quickly.

Custom create

Leverage powerful configuration options to deploy a custom solution.

Design an endpoint with multiple domains and origin groups. Define routes to connect them, and add WAF policies to protect them.

[Continue to create a Front Door](#) [Give feedback](#)

2.5 Under the Create a Front Door profile page, enter the following details in the Basics tab: create a new Resource group as **rg01** and enter the Name as **profile1**, then click on **Next: Secrets >**

Home > Create a resource > Marketplace > Front Door and CDN profiles >

Create a Front Door profile

Microsoft Azure

Basics Secrets Endpoint Tags Review + create

Azure Front Door is a modern application delivery network platform providing a secure, scalable CDN, dynamic site acceleration, and global HTTP(S) load balancing for your global web applications. [Learn more](#)

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Resource group location

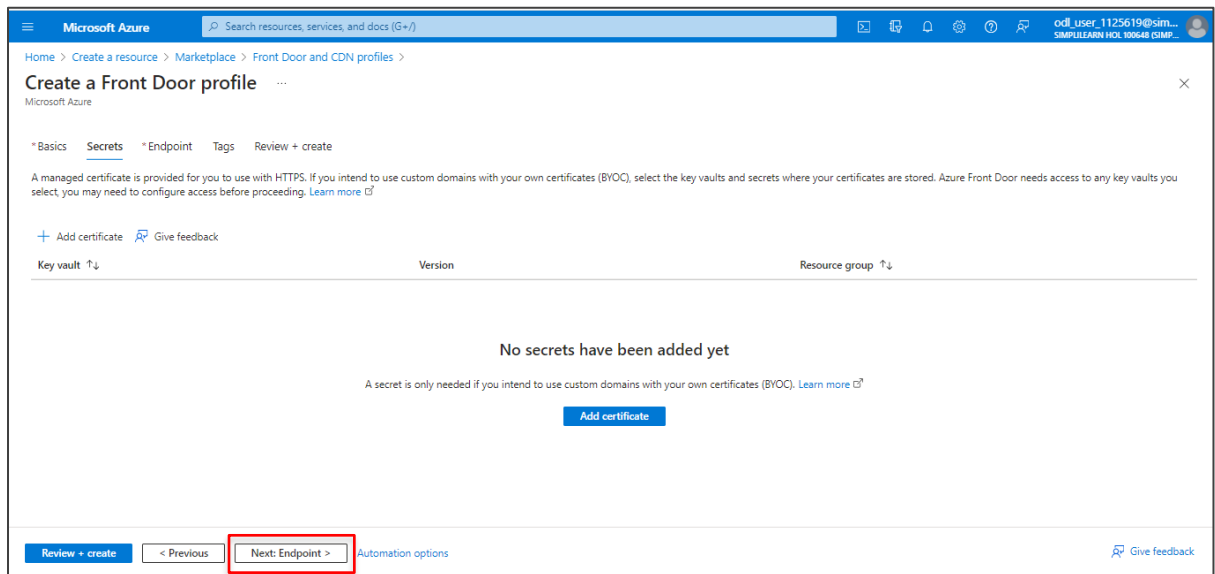
Profile details

Name *

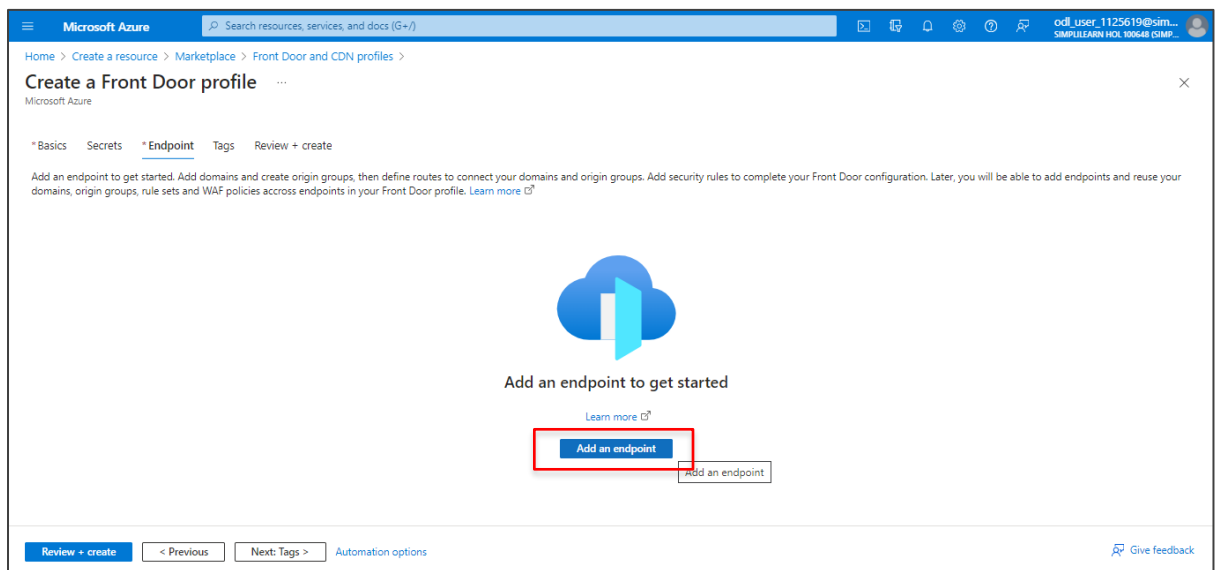
Tier * ☒ Standard Content delivery optimized ☐ Premium Security optimized

[Review + create](#) [< Previous](#) [Next: Secrets >](#) [Automation options](#) [Give feedback](#)

2.6 The page will appear as shown below. Then, click on **Next: Endpoint >**



2.7 Click on **Add an endpoint**



2.8 Under the **Add an endpoint** page, enter the **Endpoint name** as **ep1**, and click on **Add**

Microsoft Azure

Home > Create a resource > Marketplace > Front Door and CDN profiles >

Create a Front Door profile

Microsoft Azure

* Basics Secrets * **Endpoint** Tags Review + create

Add an endpoint to get started. Add domains and create origin groups, then define routes to connect your domains and origin groups. Add security rules to o domains, origin groups, rule sets and WAF policies across endpoints in your Front Door profile. [Learn more](#)

Add an endpoint to get started

[Learn more](#)

Add an endpoint

Add an endpoint

Microsoft Azure

The endpoint name specifies a desired subdomain on Front Door's default domain (i.e. z01.azurefd.net) to route traffic from that host via Front Door. You can optionally onboard custom domains as well. [Learn more](#)

Endpoint name * **ep1**

Endpoint hostname ep1-dcdafaa7d5hfbvgd.z01.azurefd.net

Status ☒ Enable this endpoint

Add **Cancel**

Review + create < Previous Next: Tags > Automation options

2.9 Click on **Review + create**

Microsoft Azure

Home > Create a resource > Marketplace > Front Door and CDN profiles >

Create a Front Door profile

Microsoft Azure

* Basics Secrets * **Endpoint** Tags **Review + create**

Add an endpoint to get started. Add domains and create origin groups, then define routes to connect your domains and origin groups. Add security rules to complete your Front Door configuration. Later, you will be able to add endpoints and reuse your domains, origin groups, rule sets and WAF policies across endpoints in your Front Door profile. [Learn more](#)

ep1-dcdafaa7d5hfbvgd.z01.azurefd.net

Routes + Add a route

Routes	Domains	Origin group	Status	Provisioning state
No results.				

Security policy + Add a policy

Name	Domain state
No results.	

Review + create < Previous Next: Tags > Automation options [Give feedback](#)

The page will appear as shown below:

Add a route

A route maps your domains and matching URL path patterns to a specific origin group. [Learn more](#)

Name *

Endpoint

Domains
[Add a new domain](#)

Patterns to match

Accepted protocols *

Redirect ☒ Redirect all traffic to use HTTPS

Origin group
[Add a new origin group](#)

[Add](#) [Cancel](#)

2.10 On the Add a route page, under Origin group, click on **Add a new origin group**; on the Add an origin group page, enter the Name as **og1** and click on **+ Add an origin**.

Add an origin group

An origin group is a set of origins to which Front Door load balances your client requests. [Learn more](#)

Name *

Origins
Origins are the application servers where Front Door will route your client requests. Utilize any publicly accessible application server, including App Service, Traffic Manager, Private Link, and many others. [Learn more](#)

[+ Add an origin](#)

Origin host name	Status	Priority	Weight
------------------	--------	----------	--------

Session affinity ☐ Enable session affinity

Health probes
If enabled, Front Door will send periodic requests to each of your origins to determine their proximity and health for load balancing purposes. [Learn more](#)

Status ☒ Enable health probes

Path *

Protocol * ☒ HTTP ☐ HTTPS

[Add](#) [Cancel](#) [Close](#)

2.11 Under the Add an origin page, enter the following details: Name as **og1**, Origin type as **App services**, select Host name and Origin host header, then click on **Add**.

Add an origin

Microsoft Azure

Origins are your application servers. Front Door will route your client requests to origins, based on the type, ports, priority, and weight you specify here. [Learn more](#)

← Go back to origin group

Name *

Origin type *

Host name *

Origin host header

Certificate subject name validation ⓘ ☒ Enable the validation

ⓘ This validation is required if private link is enabled. [Learn more](#)

HTTP port *

HTTPS port *

Priority * ⓘ

Weight * ⓘ

Status ☒ Enable this origin

Add **Cancel** **Close**

The given screenshot will appear as shown below:

Add an origin group

Microsoft Azure

An origin group is a set of origins to which Front Door load balances your client requests. [Learn more](#)

Name *

Origins

Origins are the application servers where Front Door will route your client requests. Utilize any publicly accessible application server, including App Service, Traffic Manager, Private Link, and many others. [Learn more](#)

+ Add an origin

Origin host name	Status	Priority	Weight
mydemowebapp11.azurewebsites.net	Enabled	1	1000

Session affinity ☐ Enable session affinity

Health probes

If enabled, Front Door will send periodic requests to each of your origins to determine their proximity and health for load balancing purposes. [Learn more](#)

Status ⓘ ☒ Enable health probes

Path *

Protocol * ⓘ ☒ HTTP

Home > Create a resource > Marketplace > Front Door and CDN profiles > Compare offerings > Create a Front Door profile

Add a route

Microsoft Azure

Accepted protocols * HTTP and HTTPS

Redirect ☒ Redirect all traffic to use HTTPS

Origin group
Origin group * [Add a new origin group](#)

Origin path

Forwarding protocol * ☐ HTTP only ☐ HTTPS only ☒ Match incoming request

Caching ☐ Enable caching

Rules

After you create your Front Door profile, you can create custom rules and apply them to routes. [Learn more](#)

[Add](#) [Cancel](#)

Add an origin group

Microsoft Azure

If enabled, Front Door will send periodic requests to each of your origins to determine their proximity and health for load balancing purposes. [Learn more](#)

Status ☒ Enable health probes

Path *

Protocol * ☒ HTTP ☐ HTTPS

Probe method *

Interval (in seconds) * seconds

Load balancing
Configure the load balancing settings to define what sample set we need to use to call the backend as healthy or unhealthy. The latency sensitivity with value zero (0) means always send it to the fastest available backend, else Front Door will round robin traffic between the fastest and the next fastest backends within the configured latency sensitivity. [Learn more](#)

Sample size *

Successful samples required *

Latency sensitivity (in milliseconds) * milliseconds

[Add](#) [Cancel](#) [Close](#)

2.12 Click on Add

Home > Create a resource > Marketplace > Front Door and CDN profiles > Compare offerings > Create a Front Door profile

Add a route

Microsoft Azure

A route maps your domains and matching URL path patterns to a specific origin group. [Learn more](#)

Name *

Endpoint

Domains
Domains * [Add a new domain](#)

Patterns to match

Accepted protocols * HTTP and HTTPS

Redirect ☒ Redirect all traffic to use HTTPS

Origin group
Origin group * [Add a new origin group](#)

Home > Create a resource > Marketplace > Front Door and CDN profiles > Compare offerings > Create a Front Door profile >

Add a route

Microsoft Azure

Accepted protocols * HTTP and HTTPS

Redirect ☒ Redirect all traffic to use HTTPS

Origin group og1
Add a new origin group

Origin path

Forwarding protocol *
☐ HTTP only
☐ HTTPS only
☒ Match incoming request

Caching ☐ Enable caching

Rules

After you create your Front Door profile, you can create custom rules and apply them to routes. [Learn more](#)

Add Cancel

2.13 Click on **Review + create**

Home > Create a resource > Marketplace > Front Door and CDN profiles > Compare offerings >

Create a Front Door profile

Microsoft Azure

* Basics Secrets *** Endpoint** Tags Review + create

Add an endpoint to get started. Add domains and create origin groups, then define routes to connect your domains and origin groups. Add security rules to complete your Front Door configuration. Later, you will be able to add endpoints and reuse your domains, origin groups, rule sets and WAF policies across endpoints in your Front Door profile. [Learn more](#)

ep1-dcdafaa7d5hfbxgd.z01.azurefd.net

Routes	Domains	Origin group	Status	Provisioning state
r1	1 selected	og1	Enabled	...

Security policy

Name	Domain state
No results.	

Review + create < Previous Next: Tags > Automation options

[Give feedback](#)

2.14 Click on **Create**

Home > Create a resource > Marketplace > Front Door and CDN profiles > Compare offerings >

Create a Front Door profile

Microsoft Azure

Validation passed.

* Basics Secrets * Endpoint Tags **Review + create**

Basics

Subscription: Simplilearn HOL 100648
 Resource group: rg01
 Location: East US
 Name: profile1
 Tier: Azure Front Door Standard

Endpoint

Endpoint hostname: ep1-dcdafaa7d5hfbxgd.z01.azurefd.net
 Custom domain: None
 Origin group: og1
 Origin: mydemowebapp1.azurewebsites.net
 Route: r1
 Caching: Disabled
 Security policy: None

Create < Previous Next Automation options

[Give feedback](#)

The output appears as shown below:

Microsoft Azure

Search resources, services, and docs (G+)

odl_user_1125619@sim...
SIMPLILEARN HOL 100648 (SIMP...)

Home >

profile1

Front Door and CDN profile

Purge cache Origin response timeout Delete Refresh Give feedback

Upgrade to AFD Premium to use the latest managed rulesets (argued by MS Threat Intelligence) with increased coverage, patches for specific vulnerabilities, and better false positive reduction. [Learn more](#)

Essentials [JSON View](#)

Resource group (move): [rg01](#) Name: profile1
 Status: Active Pricing Tier: Azure Front Door Standard
 Location: Global Front Door ID: a2e94c65-27a0-4b7e-bb4c-789a7a3e3ec1
 Subscription (move): [Simplilearn HOL 100648](#) Origin response timeout: 60 Seconds
 Subscription ID: 53966711-b37e-4624-8238-97b943b04221
 Tags (edit): [Add tags](#)

Properties Monitoring Recommendations

Endpoints **Custom domains**

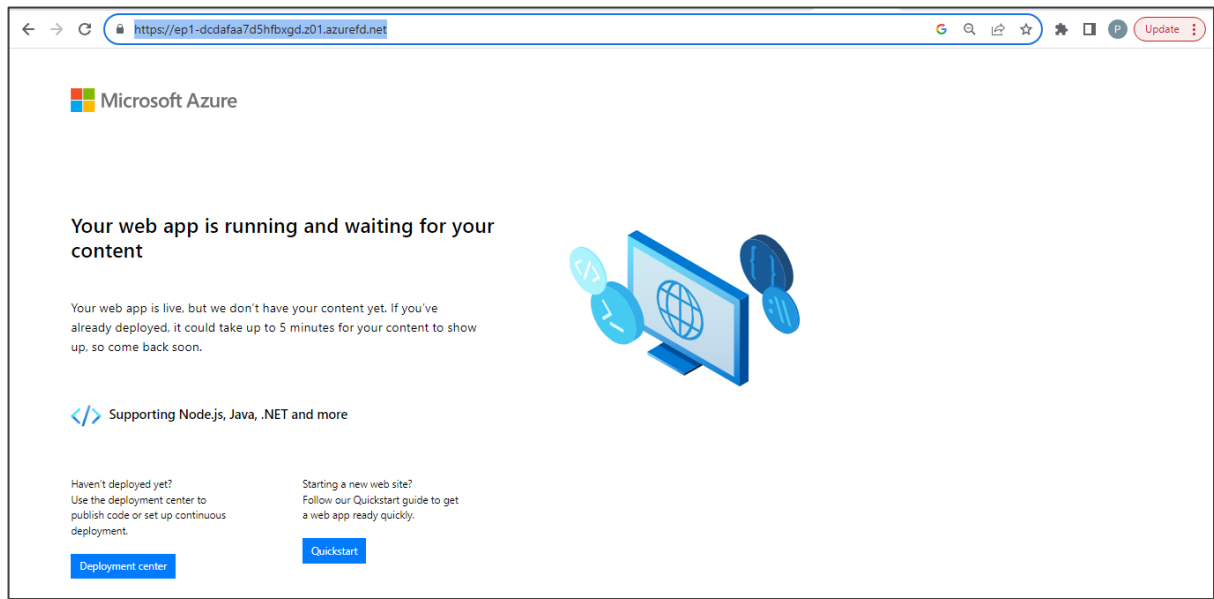
Endpoint hostname: [ep1-dcdafaa7d5hfbxgd.z01.azurefd.net](#)
 Provision succeeded
 Enabled

Security policy **Routes**

Route name: [r1](#)
 Provision succeeded

2.15 Once **profile1** is created, copy the endpoint hostname (step 2.10) and access it within the browser

The final output appears as follows:



By following the above steps, you have successfully created an Azure Front Door service to ensure secure and reliable worldwide access to your application.